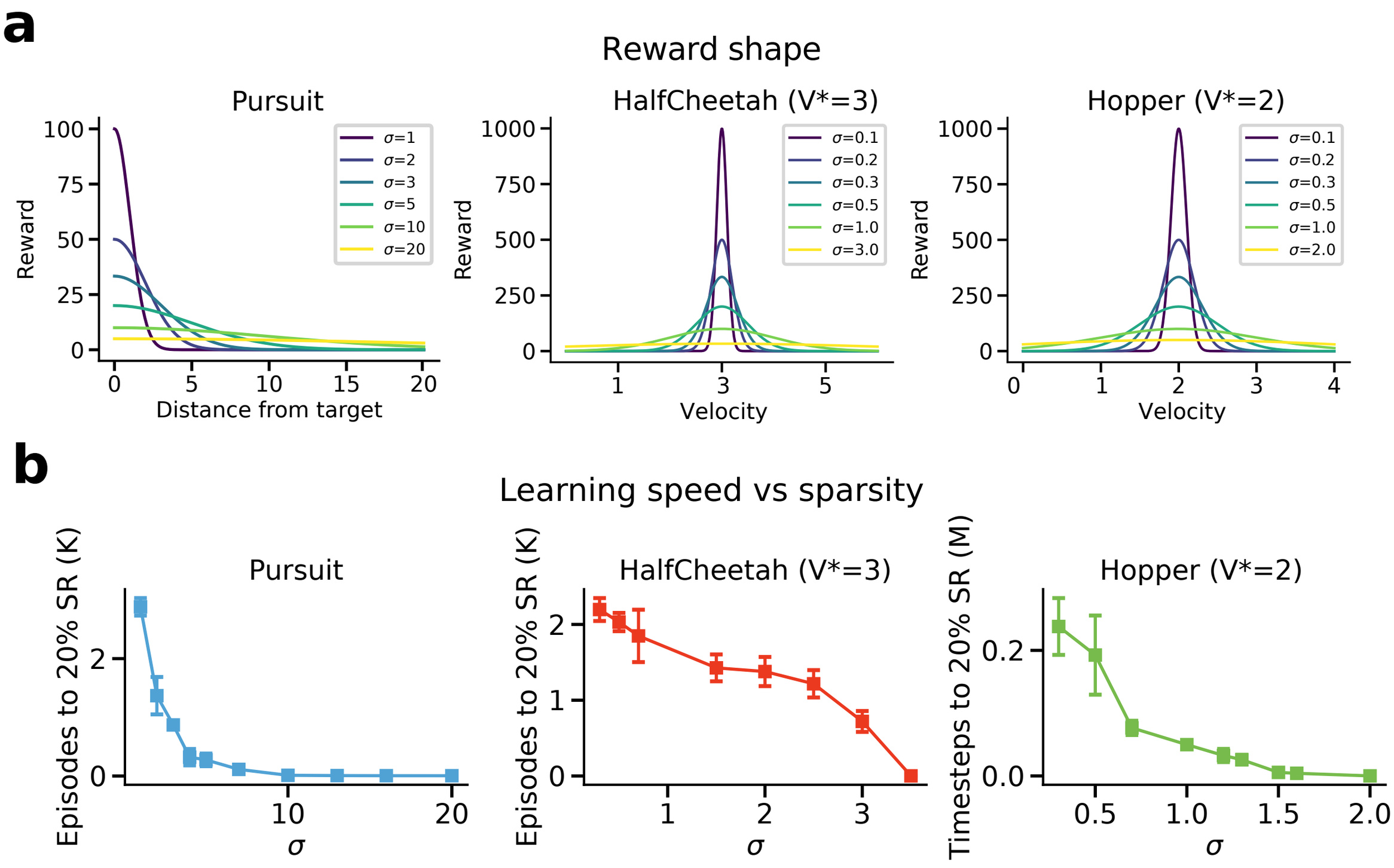

Figure 1. Effect of reward sparsity on learning speed. (A) Normalized Gaussian reward shape for different values of σ. Larger σ produces a broader reward (denser), while smaller σ concentrates reward near the target or optimal velocity (sparser). The total reward integral is conserved across all σ for the same task. For HalfCheetah, the Gaussian reward is centered at an optimal velocity of 3; for Hopper, it is centered at an optimal velocity of 2. **(B)** Learning speed as a function of σ (larger σ = denser reward). Denser rewards lead to faster learning across all three environments, with learning time increasing sharply as σ decreases. Error bars show standard error across 5 seeds.

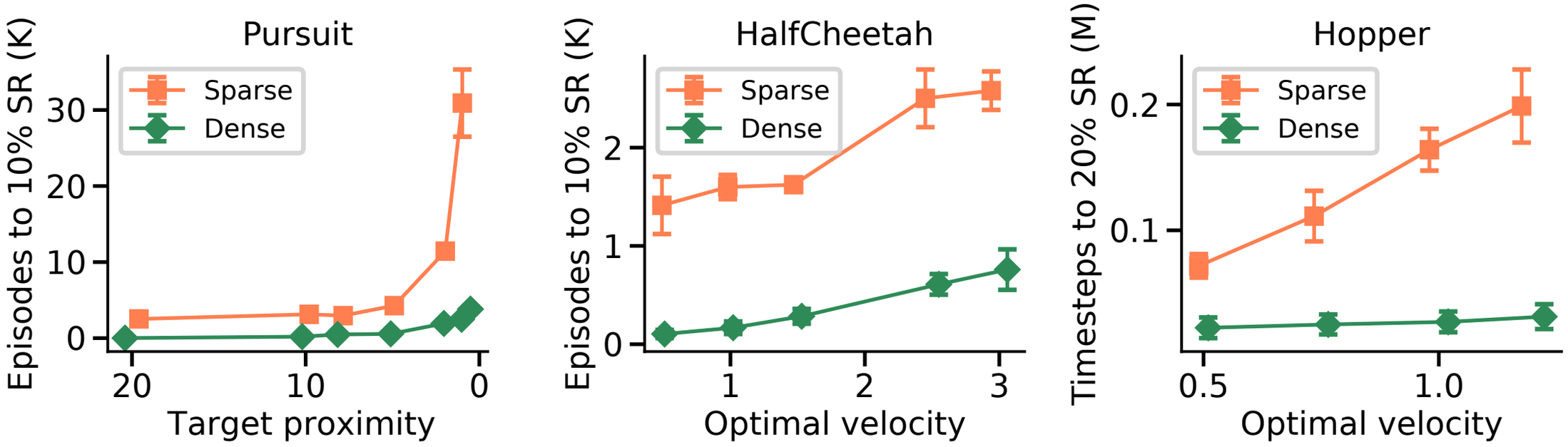

Figure 2. Scaling of learning time with goal location for sparse vs dense reward. Task difficulty is affected by the location of the Gaussian reward peak: in Pursuit, this is the distance from the target the agent must reach (smaller = harder); in HalfCheetah and Hopper, it is the optimal velocity the agent must achieve (higher = harder). The dense condition uses a σ 10 times greater than the sparse σ. Dense reward scales gracefully with increasing difficulty, while sparse reward learning time grows more steeply. Error bars show standard error across 5 seeds.

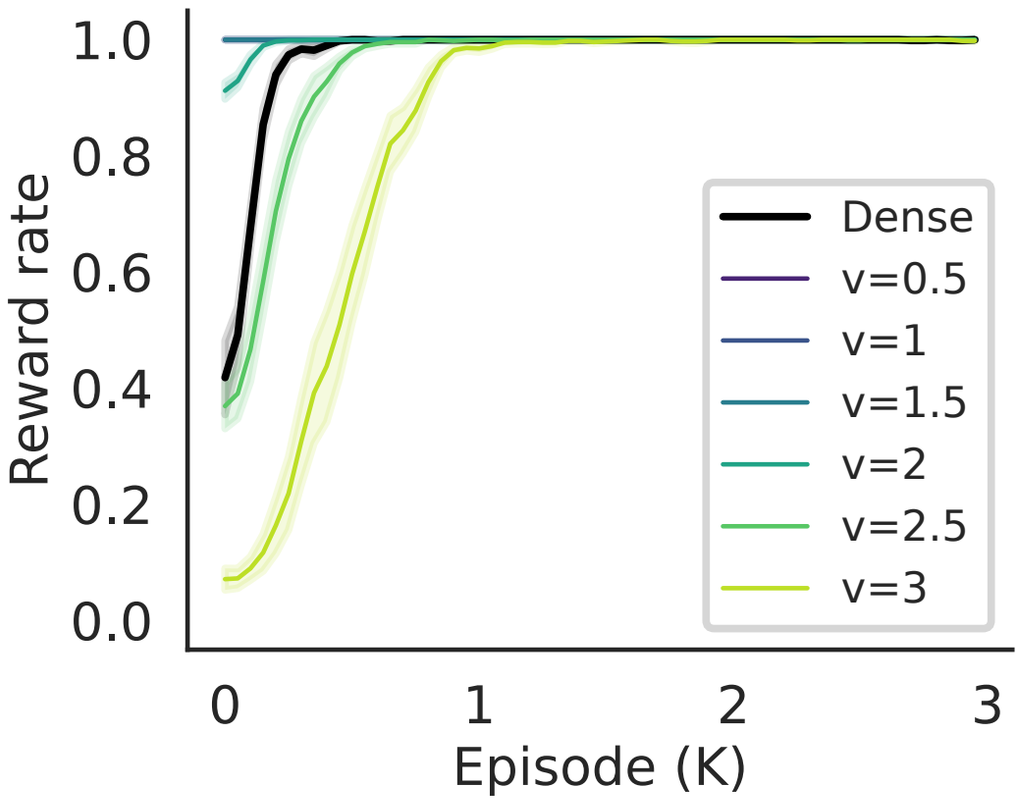

Figure 3. Fraction of episodes with positive reward in HalfCheetah. At low velocity targets (v=0.5, v=1), the sparse reward is positive on nearly 100% of episodes from the start — a random policy could exceeds these targets on some of the time steps. The dense reward (black), which equals raw forward velocity, starts lower because the agent receives negative reward when moving backward. At high velocity targets (v=2.5, v=3), the sparse reward fraction starts near zero and climbs slowly as the agent learns to reach these velocities. Shading shows standard error across seeds.